

Non-parametric tests in RStudio (Class examples: Lecture 1 to 4)

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#LECTURE 2: Wilcoxon Signed Rank Sum Test#

#Class example in R

#set working directory (file path to where R should search for your data set)

setwd("D:/UCT/STA2020F 2024/Lectures/Non-Parametric")

#read in the actual file name from the directory that you just referred to

wilcdata<-read.csv('Lecture2_Example.csv', header = TRUE, sep = ",")

#check that the data imported correctly

View(wilcdata)

#wilcox.test() will perform your required test.

#specify whether the data is paired (TRUE) or not (FALSE)

#specify exact = TRUE ONLY if you have NO TIES; in the presence of ties: exact = FALSE will give you an approximated p-value (which you want in this case)

#alternative = c("two.sided", "less", "greater")--- choose one based on the problem and alternative hypothesis that you specified

#search in the help function for more info on wilcox.test()

#NB: the value V in the output is the sum of the POSITIVE signed ranks. To obtain T, you will need to subtract the negative signed ranks yourself.

wilcox.test(x=wilcdata\$Normal,y=wilcdata\$Flexi,exact=FALSE, paired = TRUE, alternative = "greater")

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#LECTURE 3: Mann-Whitney Wilcoxon Test#

#Class example 1 (small sample approach)

datproc<-c(70,52,46,65,60,40)

typing<-c(59,70,75,85,50,82,64)

#run the wilcox.test() for the two variables

#in this case, the dat is NOT paired; let paired = FALSE

#change the alternative hypothesis to "two.sided"

wilcox.test(x=datproc,y=typing,exact=FALSE, paired = FALSE, alternative = "two.sided")

#Class example 2 (large sample approach)

#read in the data set

aspdata<-read.csv('Lectur3_Example.csv', header = TRUE, sep = ",")

#check that the data imported correctly

View(aspdata)

#run the wilcox.test() for the two variables

#in this case, the dat is NOT paired; let paired = FALSE

#change the alternative hypothesis to "greater"

```
wilcox.test(x=aspdata$New,y=aspdata$Aspirin,exact=FALSE, paired = FALSE, alternative = "greater")
```

#NB: the test statistic W (in both examples) is supposedly whichever is smaller between T1 and T2.

#In this case, R uses a different formula to perform this calculation and it is not the same as

#what we calculated it to be by hand. We will focus on the p-value and the interpretation of this test;

#the conclusions for R's test and our hand calculated tests will be the same and the p-values

#are not expected to differ by much.

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#LECTURE 4: Kruskal-Wallis Test#

#Class example 1 (equal sample sizes)

#importing data from csv file into R

```
kw1<-read.csv("Lecture4_Example1.csv", header = TRUE, sep = ",")
```

```
View(kw1)
```

#carry out the test with a formula of the columns you wish to compare: response ~ group

```
kruskal.test(formula = num ~ name, data = kw1)
```

#Class example 2 (unequal sample sizes)

#importing data from csv file into R

```
kw2<-read.csv("Lecture4_Example2.csv", header = TRUE, sep = ",")
```

```
View(kw2)
```

#carry out the test with a formula of the columns you wish to compare: response ~ group

```
kruskal.test(formula = num ~ name, data = kw2)
```

#NB: the chi-squared test statistic (in both examples) uses a different formula to perform

#this calculation and it is not the same as what we calculated it to be by hand. We will focus

#on the p-value and the interpretation of this test;

#the conclusions for R's test and our hand calculated tests will be the same and the p-values

#are not expected to differ by much.